

Single-Phase Machine

A single-phase induction motor is an electric motor that operates on a single-phase AC power supply and is widely used in domestic and light commercial applications due to its simple, robust, and cost-effective design.

Construction

A single-phase induction motor consists of two main parts: a stationary stator and a rotating rotor.

Stator: This is the stationary part, made of laminated steel to reduce energy losses. It contains the main winding and, in most types, an auxiliary (starting) winding, displaced by 90 electrical degrees from the main winding.

Rotor: The rotor is typically a squirrel-cage design, consisting of conductive bars (made of aluminum or copper) embedded in a laminated core and short-circuited by end rings. The rotor bars are often skewed to help prevent magnetic locking and ensure quieter, smoother operation.

Other components: Bearings support the rotor shaft, and some motors include a centrifugal switch or capacitor as part of the starting mechanism.



Working Principle

When a single-phase AC supply is applied to the stator's main winding, it produces a pulsating magnetic field, not a naturally rotating one. According to the double revolving field theory, this pulsating field can be seen as two equal and oppositely rotating magnetic fields, which cancel each other out at a standstill, resulting in zero net starting torque.

To **become self-starting**, the motor must employ an additional mechanism to create a phase difference in the magnetic field, simulating a rotating magnetic field during startup. This is commonly achieved using an **auxiliary winding** in conjunction with a capacitor or high resistance, which causes the current in the

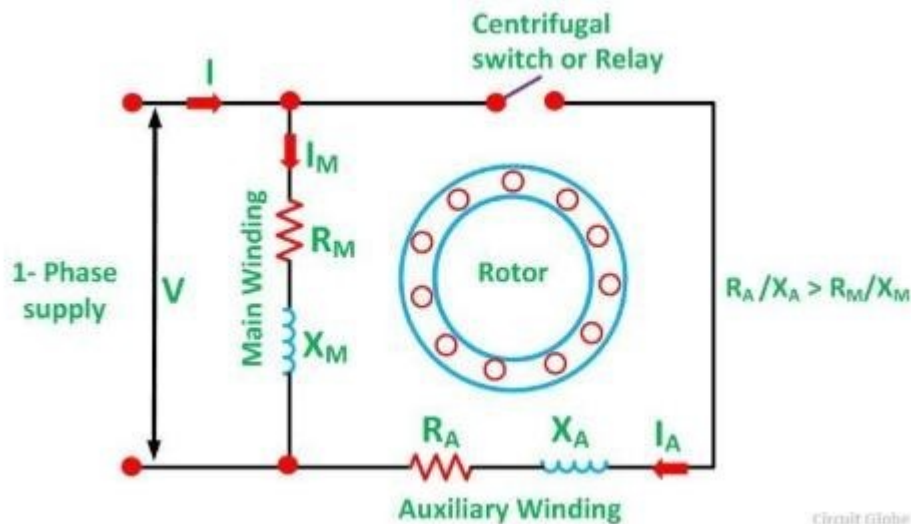
auxiliary winding to be out of phase with the current in the main winding. Once the motor reaches a sufficient speed (typically 75–80% of synchronous speed), a centrifugal switch often disconnects the auxiliary winding and starting components. The motor then continues to run on the main winding alone, relying on the induced current in the rotor to generate the necessary torque.



Types of Single-Phase Induction Motors

Single-phase induction motors are classified based on their starting methods:

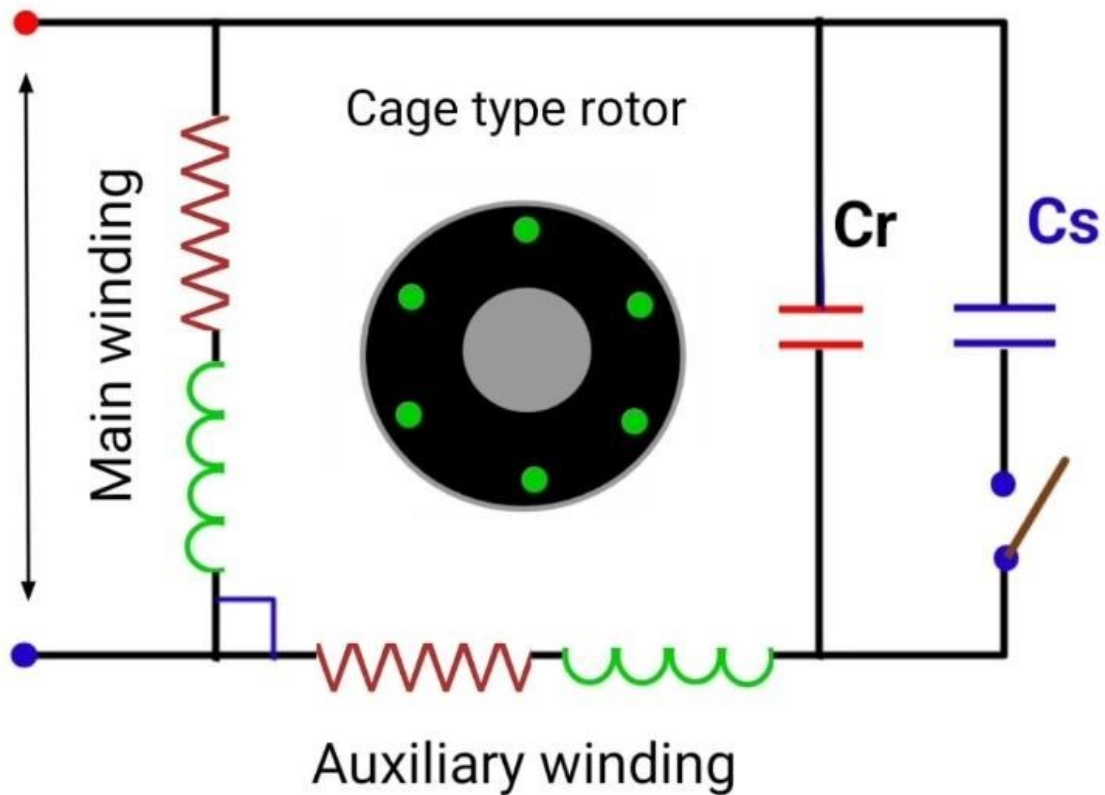
Split-Phase (Resistance Start) Motor: Uses a high-resistance auxiliary winding to create a small phase difference. They have low starting torque and are used in fans and small pumps.



Capacitor-Start Motor: Adds a capacitor in series with the auxiliary winding to produce a larger phase shift, resulting in high starting torque, making them suitable for compressors and large pumps.

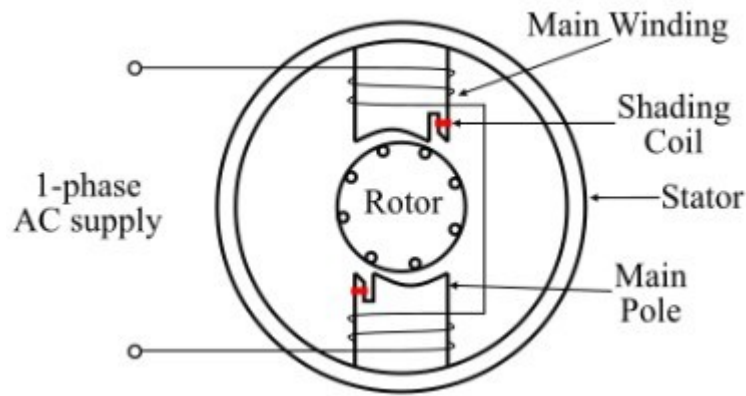
Capacitor-Start Capacitor-Run Motor: Uses two capacitors (one for starting, one for running) to offer both high starting torque and

improved running efficiency and power factor. They are used in air conditioners and high-duty blowers.



Permanent-Split Capacitor (PSC) Motor: Features a single, permanently connected capacitor, eliminating the need for a centrifugal switch. This provides smooth, quiet operation and moderate starting torque for continuous-duty applications like ceiling fans.

Shaded-Pole Motor: The simplest and least expensive type, using a short-circuited copper band (shading coil) on the poles to create a weak rotating field. They have very low starting torque and are used in small fans and toys.



Applications

Single-phase induction motors are widely used in residential, commercial, and light industrial settings where a three-phase power supply is unavailable.

Household Appliances: Fans (ceiling, table, exhaust), refrigerators, washing machines, vacuum cleaners, mixers, and hair dryers.

Commercial & Industrial: Water pumps, air conditioners, small air compressors, blowers, and various machine tools like drilling machines and lathes.

Advantages and Disadvantages

Advantages:

- Simple and robust construction, resulting in lower cost and easier maintenance.
- Compatible with standard single-phase power supplies widely available in homes and offices.

Disadvantages:

- Not self-starting (requires auxiliary mechanisms).
- Lower efficiency and power factor compared to three-phase motors.
- Lower starting torque and limited power range, generally available in fractional-kilowatt sizes.

Universal Motor

Single-phase AC series motor is a Universal Motor, characterized by its ability to run on both AC and DC power due to series-connected field and armature windings, offering high speed and torque for tools, vacuums, and mixers.



Key Characteristics & Working Principle

- **Series Connection:** Field windings are in series with the armature, similar to a DC series motor.
- **AC/DC Versatility:** Reversing current in both field and armature simultaneously results in unidirectional torque, allowing operation on AC or DC.
- **High Speed, High Torque:** Provides high starting torque and operates at high speeds, ideal for portable appliances.

- Commutation: Uses brushes and a commutator, requiring careful brush placement for optimal AC performance.
- Construction: Stator/yoke is laminated to reduce eddy current losses, and some have compensating windings for better AC performance.

Single-Phase Motor Types:

1. Repulsion Motor

A repulsion motor is a single-phase AC motor where torque is produced by repulsion between stator field and rotor.

Construction:

- Stator with single-phase winding
- Rotor similar to DC armature
- Brushes short-circuited
- No external supply to rotor

Working Principle:

- AC supply → alternating magnetic field
- Induced rotor current interacts with stator field → repulsion torque

Characteristics:

- Very high starting torque
- Variable speed
- Low efficiency

Applications: Printing presses, compressors, woodworking tools

2. Single-Phase Synchronous Motor

A synchronous motor runs at synchronous speed ($N_s = 120f/P$).

Construction:

- Stator with single-phase winding
- Rotor: PM or DC-excited
- Auxiliary winding for starting



Working Principle:

1. Starts as induction motor
2. Rotor locks to rotating magnetic field
3. Runs at synchronous speed

Characteristics:

- Zero slip
- Low starting torque
- High efficiency

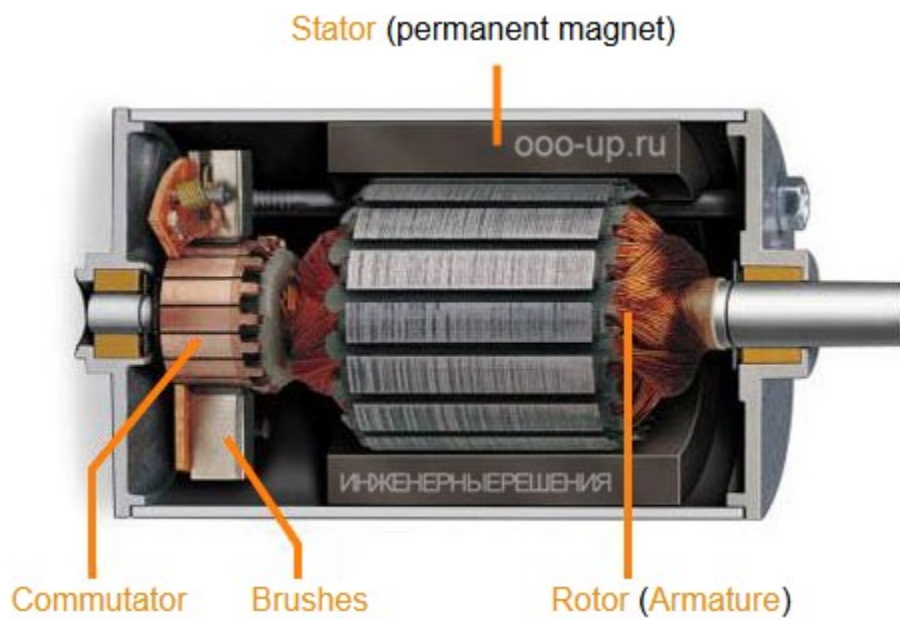
Applications: Clocks, timers, servo systems

3. Brushed DC Motor

A brushed DC motor converts DC to mechanical energy using brushes and commutator.

Construction:

- Stator: magnets
- Rotor: armature windings
- Brushes & commutator



Working:

- DC flows through brushes → armature
- Produces magnetic torque
- Commutator reverses current automatically

Characteristics:

- High starting torque
- Easy speed control

- Requires brush maintenance

Applications: Toys, drills, wipers, small appliances

4. Brushless DC Motor (BLDC)

BLDC uses electronic commutation (no brushes).

Construction:

- Stator: 3-phase windings
- Rotor: permanent magnets
- Controller with Hall sensors

Working:

- Controller energizes stator phases in sequence
- Produces rotating magnetic field
- Rotor follows field

Characteristics:

- High efficiency
- Durable
- Quiet operation

Applications: Drones, robotics, EVs, HDD

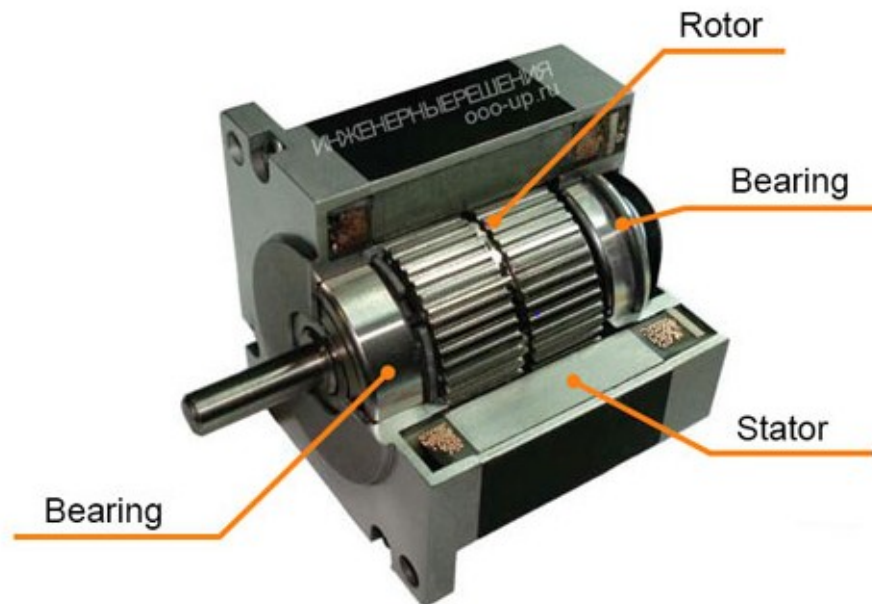
Special types of motor

1. Single-Phase Stepper Motor

A **stepper motor** converts electrical pulses into **precise rotational steps**.

Key Features

- Operates on **single-phase AC or DC pulses**.
- Rotor moves in **discrete steps**.
- Suitable for **position control** applications.



How It Works

- Stator has **poles** energized in a sequence.
- Rotor is usually **permanent magnet or variable-reluctance** type.
- Each pulse → rotor shifts by a **fixed angle** (e.g., 1.8° , 7.5° , etc.)

Advantages

- Precise positioning
- No feedback needed (open-loop)
- Simple construction

Applications

- Printers
- CNC machines
- Robotics
- 3D printers

2. Single-Phase Hysteresis Motor

A hysteresis motor is a **synchronous motor** that uses the **hysteresis property** of magnetic materials.

Key Features

- Starts as an **induction motor**
- Runs at **synchronous speed**
- Smooth and quiet operation

Construction

- Stator: Single-phase winding
- Rotor: Solid cylinder of **high hysteresis alloy** (e.g., cobalt steel)
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Operation

1. **Starting:**

- The rotor lags the rotating magnetic field → produces **induction torque**.

2. **Running:**

- Rotor locks into synchronism because of **hysteresis torque**.
- No teeth → no cogging → very smooth.

Advantages

- Quiet
- High starting torque
- Constant speed under load

Applications

- Clocks
- Timers
- Recorders
- Precision instruments